

Pritom Kumar Das

Doctoral Researcher · Human–Computer Interaction · Agentic AI · Software Engineer

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Doctoral researcher in Human–Computer Interaction at the University of Helsinki, funded by Finland's national AI doctoral programme (FCAI AI-DOC). Studies how autonomous AI agents communicate risk to the people supervising them, combining interviews and controlled comparison studies with hands-on engineering. Background in software engineering (BSc, University of Dhaka) and data science (MSc, University of Helsinki); three peer-reviewed publications.

DOCTORAL RESEARCH

Doctoral Researcher, Human–Computer Interaction

Aug 2024 – Present

Ubiquitous Interaction Research Group, Department of Computer Science, University of Helsinki

Primary supervisor: Prof. Giulio Jacucci · Secondary supervisors: Michael Mathioudakis, Robin Welsch · Funded by FCAI AI-DOC

- Designing and evaluating risk-calibrated permission notifications for agentic AI coding tools: developer interviews, a designed notification, and a comparison study against two alternatives (no signal; a blunt rule that flags every deletion).
- Building and testing a pipeline that gives embodied AI agents conversational capability in VR; researching LLM fine-tuning and sparse autoencoders.
- Participant in *MyModel: Interactive Models for Resourceful Recommendations* (Academy of Finland, 2023–2027).

PUBLICATIONS

ACM TOCHI First-author manuscript for ACM Transactions on Computer–Human Interaction

First author · title withheld pending anonymous review

ACM TOIS 2025 Incorporating Cognitive Abilities into Web Search Re-ranking

Tung Vuong, **Pritom Kumar Das**, Tuukka Ruotsalo · ACM Transactions on Information Systems 43(5), 1–35 · doi:10.1145/3736401

IEEE VR 2025 Immersive Tailoring of Embodied Agents Using Large Language Models

Andrea Bellucci, Giulio Jacucci, Kien Duong Trung, **Pritom Kumar Das**, Sergei Viktorovich Smirnov, Imtiaj Ahmed, Jean-Luc Lugrin · IEEE Conference on Virtual Reality and 3D User Interfaces, 392–400 · doi:10.1109/VR59515.2025.00063

JASIST 2024 Associating cognitive abilities with naturalistic search behavior

Tung Vuong, **Pritom Kumar Das**, Tuukka Ruotsalo · Journal of the Association for Information Science and Technology 76(4), 665–685 · doi:10.1002/asi.24963

EDUCATION

MSc, Data Science

Sep 2022 – Jul 2024

University of Helsinki, Finland · full scholarship (100%)

Thesis: “*Cognition-aware Dwell Time Modeling: Evaluating Relevance and Usefulness*” · exploratory data analysis, data visualisation, research procedure, machine learning, large language models

BSc, Software Engineering

Jan 2017 – Aug 2021

Institute of Information Technology, University of Dhaka, Bangladesh

Final-year project: *EarnEasy*, a crowdsourcing app · software development and testing, algorithms and data structures, mobile and web development

RESEARCH & INDUSTRY EXPERIENCE

Research Assistant

Jan – Jul 2024

Ubiquitous Interaction Research Group, University of Helsinki · Supervisor: Prof. Giulio Jacucci

- Built and tested software for user context and interaction logging; analysed user context data with LLMs (human–computer interaction, information retrieval, user modelling).

Research Assistant

Feb – Dec 2023

Cognitive Computing Research Group, University of Helsinki · Supervisor: Tuukka Ruotsalo

- Analysed and modelled participants' cognitive-ability data with statistical testing, regression analysis and ablation tests; used LLMs to find semantic similarity between user tasks.
- Tools: Jupyter Notebook, TensorFlow, Hugging Face.

Software Engineering Intern

Jan – Jun 2020

SELISE, Dhaka, Bangladesh · Swiss-based software company · Health Care Team of nine developers

- Backend development in the .NET framework; transformed legacy Visual Basic software to .NET with Entity Framework; built desktop forms and dashboards in an agile team.
- Awarded “Sprint Hero” as the team's most outstanding intern.

TECHNICAL SKILLS

RESEARCH: experimental design, controlled studies, qualitative interviews, statistical and regression analysis, ablation testing

LANGUAGES: Python, C#, Java, C++, Dart, TypeScript, SQL

ML / AI: PyTorch, TensorFlow, Keras, Hugging Face, LLM fine-tuning, sparse autoencoders, speech tools (Whisper, pyannote), OpenCV, MediaPipe, embodied agents, VR

ENGINEERING: .NET, Entity Framework, Flutter, Django, Docker, Git, Jira

FUNDING, TEACHING & RECOGNITION

- **Funding:** FCAI AI-DOC doctoral funding (Language and Speech Technology); full (100%) scholarship for the MSc in Data Science.
- **Teaching assistant,** Introduction to Data Science and Introduction to Artificial Intelligence, University of Helsinki: grading, exercise sessions, guiding students.
- **Service:** Student Representative, Data Science Master's Programme steering group (Jan 2023 – Aug 2024); International Student Tutor (Aug – Dec 2023).
- AHZ Associates' Special Achievement Award, outstanding bachelor's performance (2021)
- The Duke of Edinburgh's International Award: Gold (2022), Silver (2021), Bronze (2019)
- Runner-up, Dash Hackathon (2022)
- 1st Runner-Up, SPL Project Showcasing, University of Dhaka (2019)
- 3rd position, ICT Quiz, Independence Cup, IIT, University of Dhaka

LANGUAGES

English — C2 (Proficiency) · Bengali — native